SCCM Basics – Software Installation and Updates

SCCM (System Center Configuration Manager), now known as **Microsoft Configuration Manager**, is a systems management tool used by IT administrators to deploy software, install updates, manage devices, and ensure compliance across an organization.

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1. Software Installation in SCCM

Software installation in SCCM allows administrators to remotely install applications on one or thousands of computers.

Process Flow

1. Create the application package.
2. Define installation and uninstall commands.
3. Distribute content to Distribution Points (DPs).
4. Deploy the application to a user or device collection.
5. Client devices download and install the software.

Example

Deploying Google Chrome to 500 computers:

- Download the Chrome installer.
- Create a new Application in SCCM.
- Specify:
    - Installation command:

ChromeSetup.exe /silent /install

    - Detection Method:
        - Registry
        - MSI Product Code
        - File existence
- Distribute the application.
- Deploy to the "All Office PCs" collection.

The SCCM client installs Chrome automatically on target computers.

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2. Software Deployment Types

Common deployment types include:

- MSI Installer
- EXE Installer
- Script Installer (PowerShell, Batch, VBScript)
- App-V Virtual Applications
- Microsoft Store Applications

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3. Required vs Available Deployment

Required Deployment

- Automatically installs software.
- No user action required.
- Used for mandatory applications.

Example:

- Antivirus
- VPN Client
- Microsoft Office

Available Deployment

- Users install software from **Software Center**.
- Optional applications.

Example:

- VLC Media Player
- Adobe Reader

- Notepad++

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4. Software Center

Software Center is the user-facing application installed with the SCCM client.

Users can:

- Install approved software
- View installation status
- Retry failed installations
- Install operating systems (if enabled)
- Install updates
- View maintenance windows

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5. Application Deployment Workflow

Administrator

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Create Application

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Distribute Content

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Deploy to Collection

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Client Downloads Content

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Install Application

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Report Success/Failure

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6. Software Updates in SCCM

SCCM integrates with **Windows Server Update Services** to deploy operating system and Microsoft product updates.

Administrators can deploy:

- Windows Updates
- Microsoft Office Updates
- SQL Server Updates
- Defender Updates
- Driver Updates

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7. Software Update Process

1. Synchronize updates from Microsoft.
2. Select required updates.
3. Download update content.
4. Create a Software Update Group (SUG).
5. Deploy the update group.
6. Clients install updates.
7. Restart if required.

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8. Software Update Groups (SUG)

A Software Update Group is a collection of updates deployed together.

Example:

Monthly Patch Group

- Windows 11 Security Update
- Office Security Update
- Defender Definitions

Benefits:

- Easier management
- Simplified reporting
- Single deployment

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9. Automatic Deployment Rules (ADR)

ADRs automate update deployments.

Example:

Every **Patch Tuesday**:

- Search for new Critical Updates
- Download updates
- Create a new Software Update Group
- Deploy to Pilot Devices
- Deploy to Production after testing

Benefits:

- Saves time
- Ensures consistency
- Reduces manual effort

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10. Deployment Collections

Software and updates are targeted using collections.

Examples:

- All Workstations
- Finance Department
- HR Users
- Windows 11 Devices
- IT Test Group

Deployments can target:

- Device Collections
- User Collections

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11. Monitoring Deployments

Administrators monitor deployments through:

- Deployment Status
- Success Rate
- Error Codes
- Client Compliance
- Installation Progress

Common statuses include:

- Success
- In Progress
- Failed
- Unknown

- Requirements Not Met

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12. Common SCCM Log Files

Log File	Purpose
AppDiscovery.log	Application detection
AppEnforce.log	Application installation
CAS.log	Content download
ContentTransferManager.log	Content transfer
UpdatesDeployment.log	Update deployment
WUAHandler.log	Windows Update Agent communication
ScanAgent.log	Update scan
ExecMgr.log	Program execution

Log location:

C:\Windows\CCM\Logs

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13. Best Practices

- Test deployments on a pilot collection before production.
- Use maintenance windows to avoid disrupting users.
- Configure detection methods carefully to prevent reinstall loops.
- Keep Distribution Points healthy and synchronized.
- Monitor deployment reports and investigate failures promptly.
- Automate recurring update deployments with ADRs where appropriate.

Interview Questions

Q1. What is SCCM?
Answer: SCCM (Microsoft Configuration Manager) is an endpoint management solution used to deploy software, manage updates, inventory devices, and administer Windows systems.

Q2. What is the difference between Required and Available deployments?
Answer: Required deployments install automatically, while Available deployments allow users to install software manually from Software Center.

Q3. What is Software Center?
Answer: Software Center is the client application where users can install approved software, updates, and view deployment status.

Q4. What is a Software Update Group (SUG)?
Answer: A Software Update Group is a collection of updates managed and deployed together.

Q5. What is an Automatic Deployment Rule (ADR)?
Answer: ADR automates the process of selecting, downloading, grouping, and deploying updates based on predefined criteria.

Q6. Where are SCCM client logs stored?
Answer: The default location is:

C:\Windows\CCM\Logs

Q7. Which log is commonly used to troubleshoot application installation failures?
Answer: AppEnforce.log is the primary log for troubleshooting application installation issues.